

CTP FINAL PROJECT REPORT

Project Title: Case File-Sabotage at AUW Server Room

Student 1 Details

Name: Suvodra Chakma

ID: 242161

AUW Email: auw242161@auw.edu.bd

Student 2 Details

Name: Mittra Chakma

ID: 242313

AUW Email: auw242313@auw.edu.bd

Student 3 Details

Name: F.Lallawmzuali

ID: 242169

AUW Email: auw242169@auw.edu.bd

Date: 12th August 2026

1. Abstract

This project presents an interactive web-based detective simulation game titled “**Case File: Sabotage at AUW Server Room**”. Developed using HTML, CSS, and JavaScript, the game challenges players to solve a mystery involving three primary suspects Mittra, Lomi, and Suvodra by examining evidence on an interactive corkboard, interrogating suspects, and discovering hidden clues. The objective of the project was to create a responsive, engaging, and state-driven web experience with a smooth user interface, score-tracking mechanisms, dynamic DOM manipulation, and logical puzzle design.

2. Introduction

Modern web applications rely heavily on dynamic user interactions and real-time state management. The primary goal of this project was to design and implement a lightweight, browser-based detective narrative game centered around a fictional campus event at the Asian University for Women (AUW).

Objectives:

- Design an interactive corkboard UI for structured evidence presentation.
- Implement dynamic logic for searching hidden clues, interrogating suspects, and score deductions.
- Create a progressive difficulty curve for clues (Clue 1, Clue 2, Clue 3)
- Build a modular codebase separating layout (HTML), styling (CSS), and logic (JavaScript).

3. Methodology / Implementation

Functional UI Component Layout:

1. **Header Bar:** Displays the main title ("Case File: Sabotage at AUW Server Room") and the real-time **Detective Score** badge (initial value: 350).
2. **Left Panel (Briefing & Suspect Navigation):** Displays case background story and clickable suspect buttons featuring profile avatars and student major details.
3. **Center Panel (Interactive Corkboard):** Renders physical evidence cards with red pin icons (`.pin`). Cards expand into detailed modal dialogs upon click.
4. **Right Panel (Investigate & Accuse Actions):** Features the "**Search Room for Hidden Clues (-100 pts)**" button and the suspect accusation dropdown menu with submission trigger.

4. Results & Webpage Outcomes

The application was tested across standard browser environments to demonstrate user interactions, modal pop-ups, dynamic point deductions, and victory scenarios.

Game play Screenshots & Outcome Analysis:

Outcome 1: Initial Game Board & Interface Setup

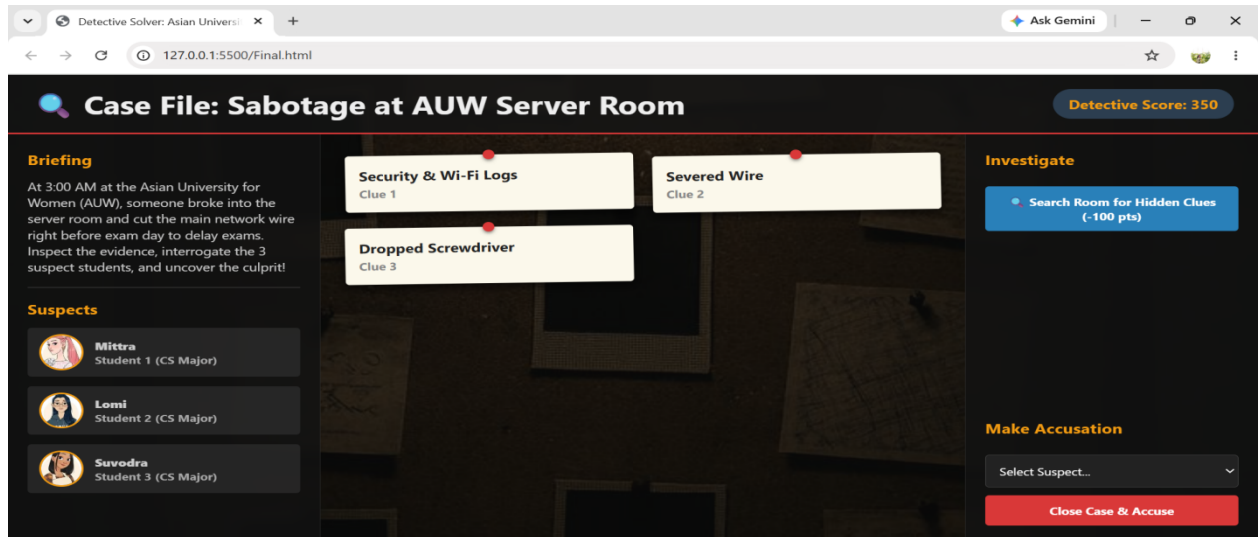


Figure 1: Main Webpage Interface upon page load.

Webpage Outcome: Displays the main layout with a starting **Detective Score of 350**. The corkboard initializes with 3 core evidence cards ("Security & Wi-Fi Logs", "Severed Wire", and "Dropped Screwdriver"). On the left, profile cards for suspects **Mittra**, **Lomi**, and **Suvodra** are rendered.

Outcome 2: Suspect Interrogations (Interactive Modals)

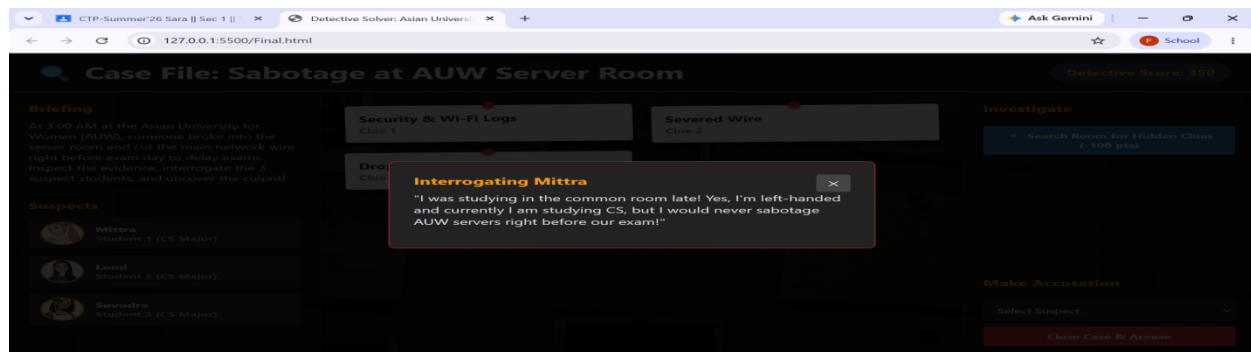


Figure 2: Interrogating Mittra Modal Pop-Up.

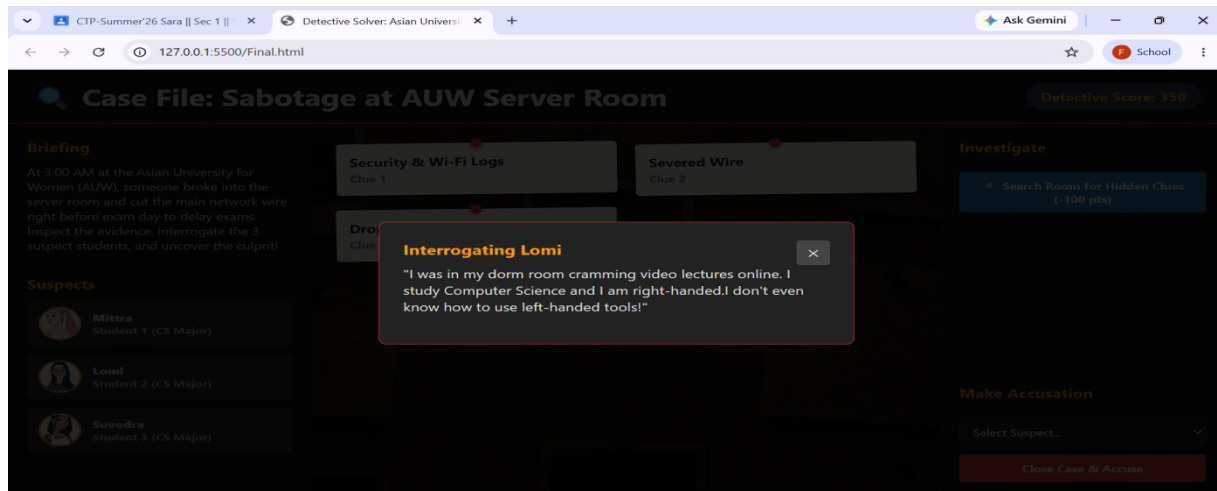


Figure 3: Interrogating Lomi Modal Pop-Up.

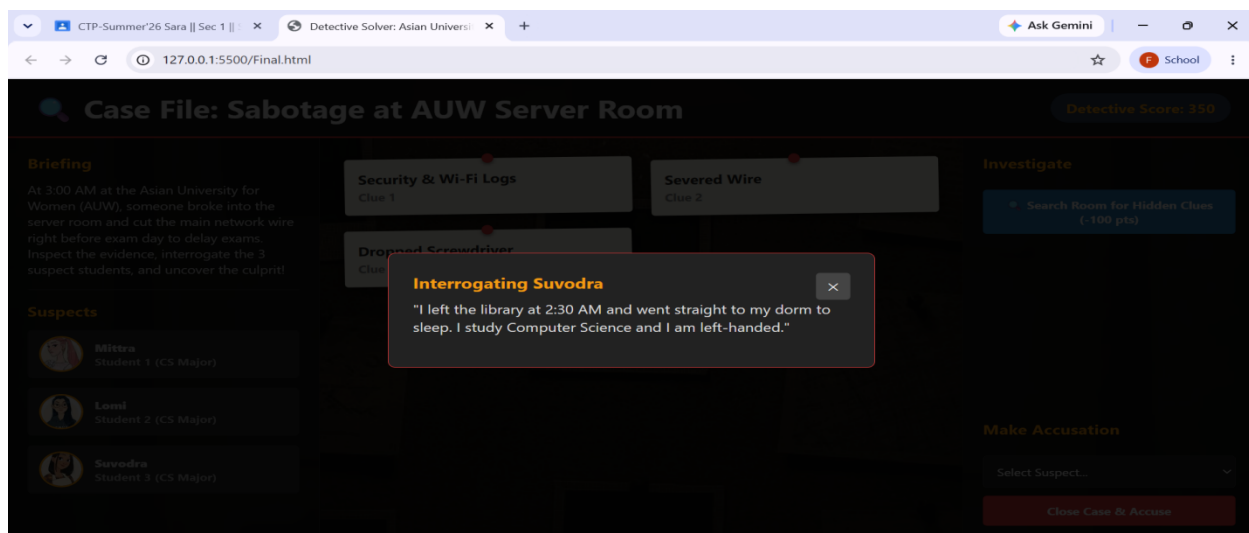


Figure 4: Interrogating Suvodra Modal Pop-Up.

Webpage Outcome: Clicking any suspect profile button on the left panel triggers a modal dialogue overlay displaying their statement and handedness:

- *Figure 2 (Mittra):* Confirms she was studying late in the common room, studies CS, and is **left-handed**.
- *Figure 3 (Lomi):* Confirms she was in her dorm watching online lectures, studies CS, and is **right-handed** (doesn't know how to use left-handed tools).
- *Figure 4 (Suvodra):* Confirms she left the library at 2:30 AM to go to her dorm, studies CS, and is **left-handed**.

Outcome 3: Standard Clue & Evidence

Figure 5: Security & Wi-Fi Logs Modal Pop-Up:

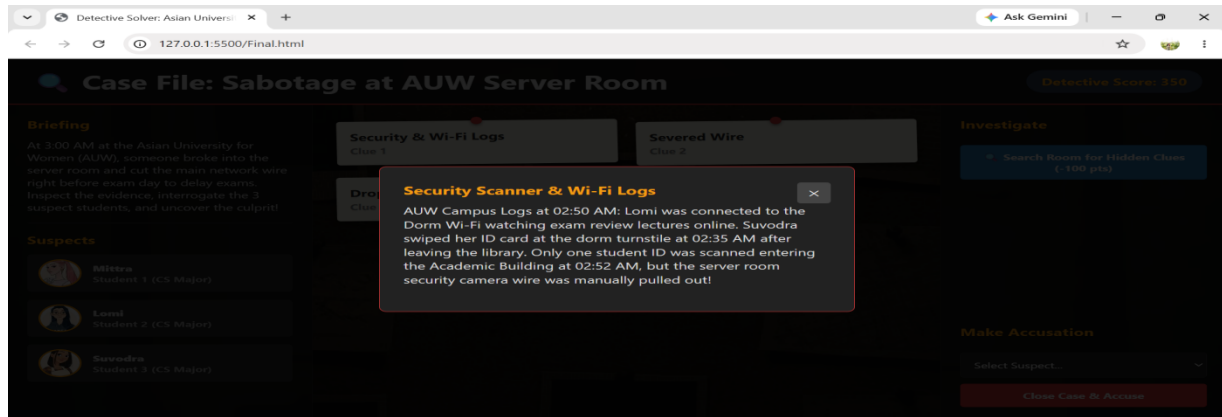


Figure 6: Severed Wire Cut Clue Modal Pop-Up:

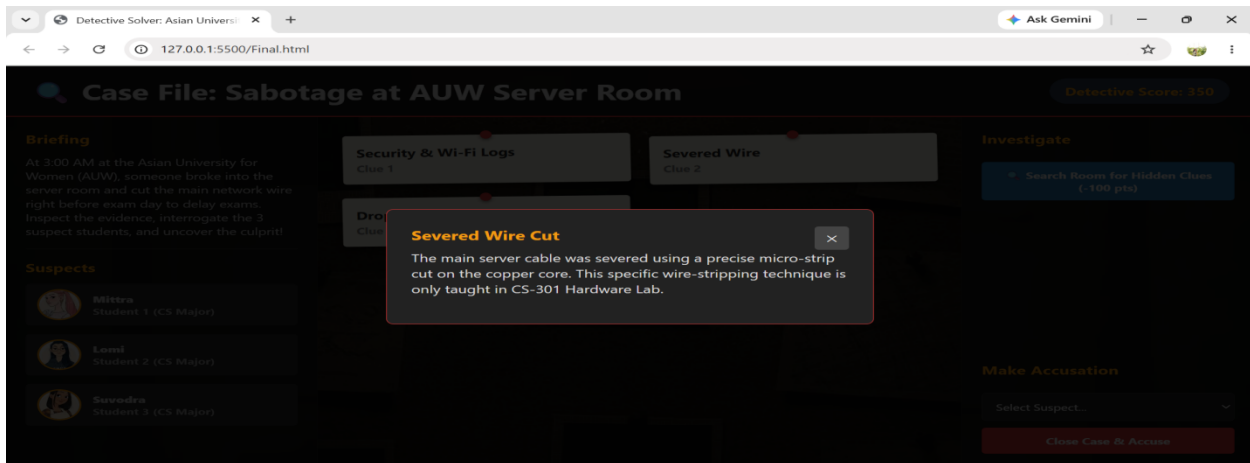
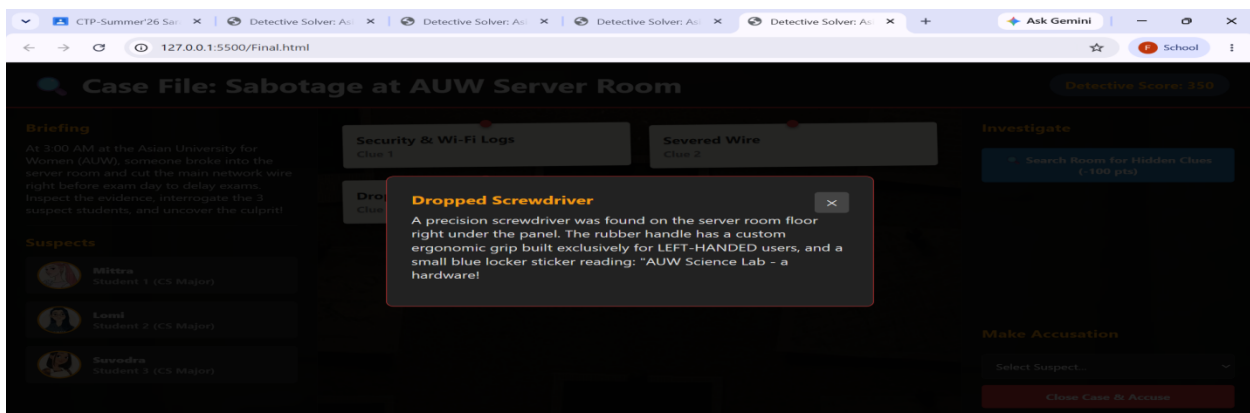


Figure 7: Dropped Screwdriver Evidence Modal Pop-Up:



Webpage Outcome: Clicking any evidence card triggers a central modal dialog over a darkened backdrop.

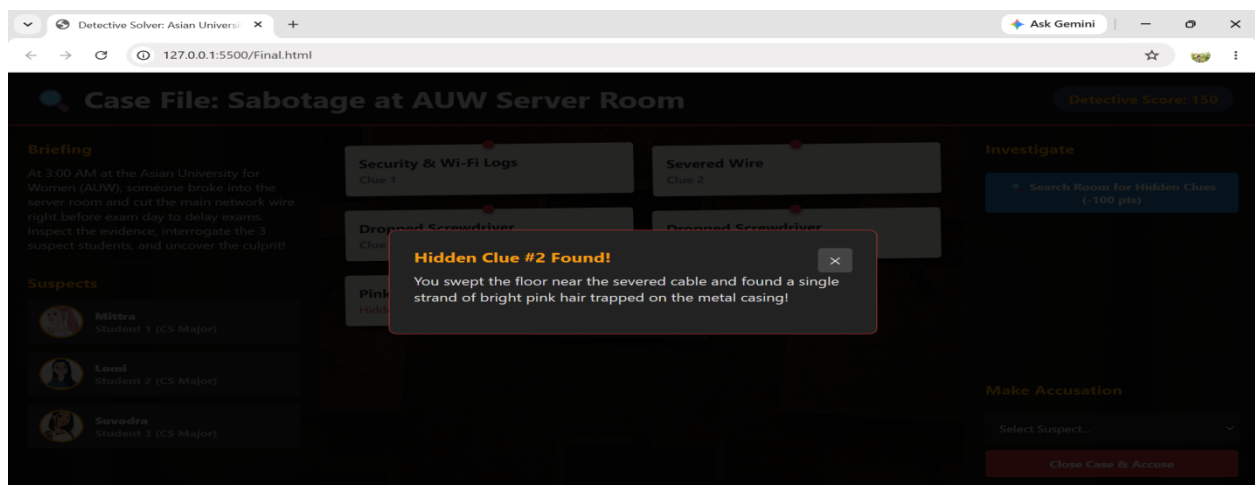
- *Figure 5* reveals security log data: Lomi watched lectures in her dorm, Suvodra swiped her ID at the turnstile at 02:35 AM, and one unknown ID swiped into the Academic Building at 02:52 AM.
- *Figure 6* reveals technical details: the server cable was cut using a micro-strip technique taught exclusively in CS-301 Hardware Lab.
- *Figure 7* shows physical evidence: a left-handed precision screwdriver labeled "A UW Science Lab – a hardware

Outcome 4: Searching Room for Hidden Clues (-100 Points Deduction)

Figure 8: Hidden Clue #1 Discovery Modal (Score drops from 350 to 250):



Figure 9: Hidden Clue #2 Discovery Modal (Score drops from 250 to 150):



Outcome 5: Searching Room for Hidden Clues (-100 Points Deduction)

Webpage Outcome: Clicking the "Search Room for Hidden Clues (-100 pts)" button executes dynamic score calculations and clue insertion:

- First Click (*Figure 8*): Deducts **100 points** (score decreases to **250**), displays a modal announcing "Hidden Clue #1 Found!", and appends the Left-Handed Screwdriver findings.
- Second Click (*Figure 9*): Deducts another **100 points** (score decreases to **150**), displays a modal announcing "Hidden Clue #2 Found!", and appends a new card showing a **bright pink hair strand** trapped on the metal casing.

Outcome 6: Dynamic DOM Injection & Total Score Verification on Corkboard

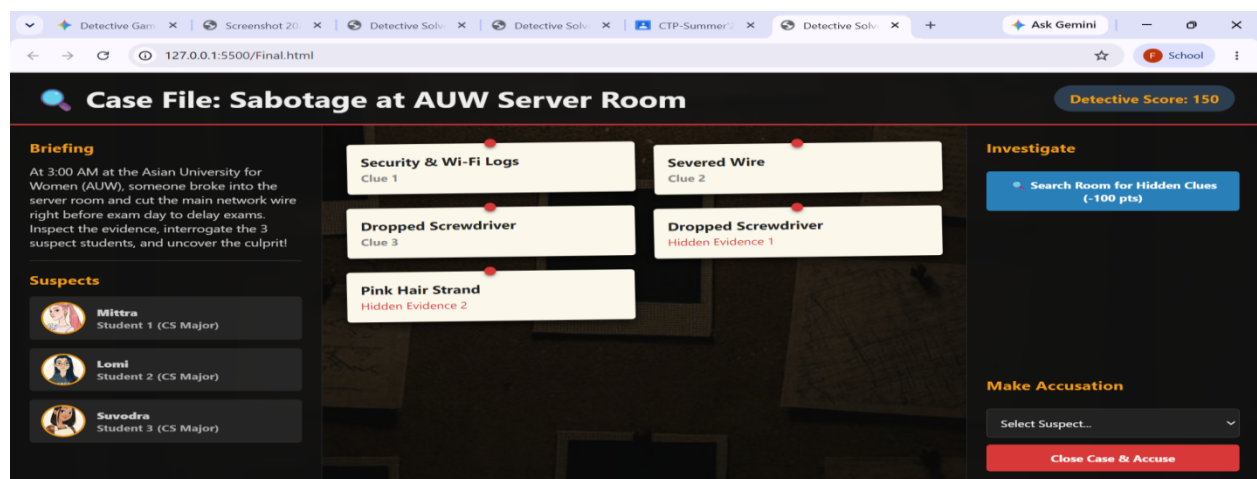


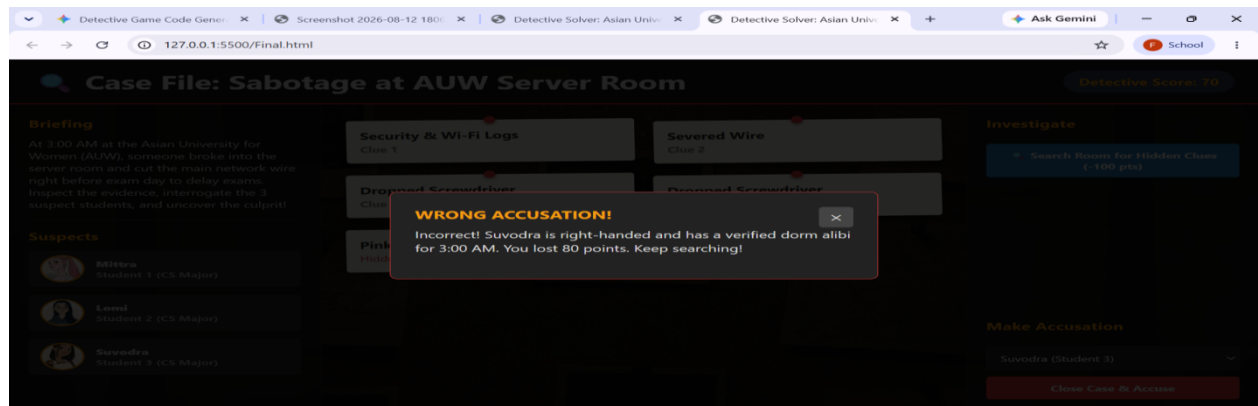
Figure 10: Main game board displaying injected hidden evidence cards and updated score of 150.

Webpage Outcome: This view confirms that the DOM dynamic injection and score state mechanics work simultaneously on the main webpage layout:

- **Dynamic Card Injection:** Both "**Dropped Screwdriver (Hidden Evidence 1)**" and "**Pink Hair Strand (Hidden Evidence 2)**" are rendered directly onto the corkboard grid alongside the initial evidence cards, featuring distinct red subheadings.
- **Real-Time Score Update:** The **Detective Score** badge in the upper-right corner reflects the accumulative penalty of both searches ($\$350 - 100 - 100\$$), displaying a final score of **150 points** prior to making an accusation

Outcome 7: Penalty Handling on Incorrect Accusation (-80 Points Deduction)

Figure 11: False Accusation Penalty Modal:



Webpage Outcome: Selecting an innocent suspect (e.g., **Suvodra**) from the dropdown menu and clicking "Close Case & Accuse" triggers a penalty outcome. A modal titled "**WRONG ACCUSATION!**" informs the player that Suvodra is right-handed and has a verified dorm alibi. The player is penalized **-80 points** (score drops from 150 to **70 points**).

Outcome 8: Successful Case Resolution & Game Victory

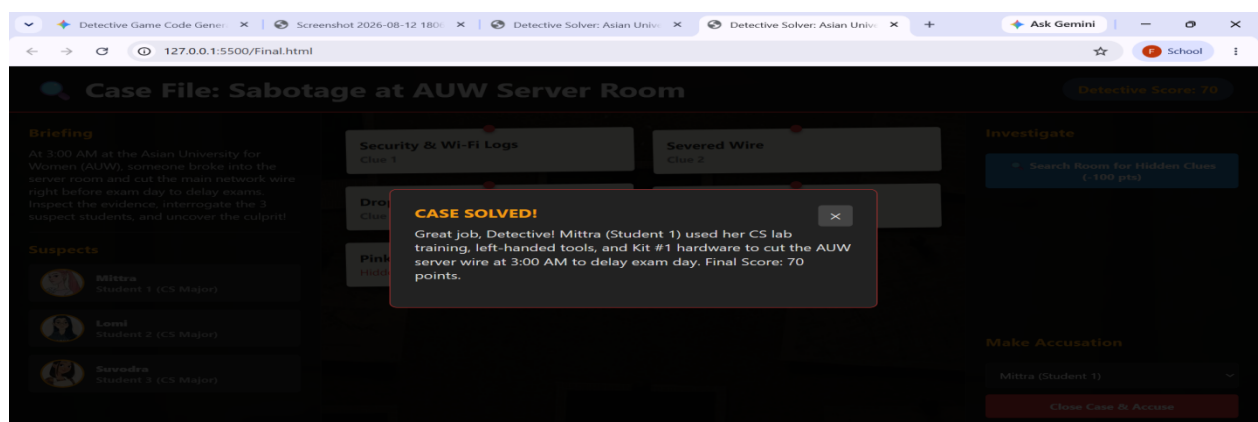


Figure 12: "CASE SOLVED!" Victory Modal.

Webpage Outcome: Selecting **Mitra** from the dropdown menu and clicking "Close Case & Accuse" triggers the winning outcome. A modal titled "**CASE SOLVED!**" pops up, verifying that Mitra used her CS lab training, left-handed tools, and Kit #1 hardware to cut the wire at 3:00 AM. It displays the final calculated score of **70 points**.

5. Challenges and Limitations

Challenges Faced:

1. **Dynamic Point Calculation:** Aligning the UI search button text (-100 pts) with background state variables to ensure that searching for hidden clues subtracts 100 points and wrong accusations subtract 80 points reliably without falling below zero.
2. **Visual Contrast & Layering:** Ensuring that dark overlay modals (.modal) render smoothly over the transparent background corkboard layer (.bg-layer) while keeping modal text readable.

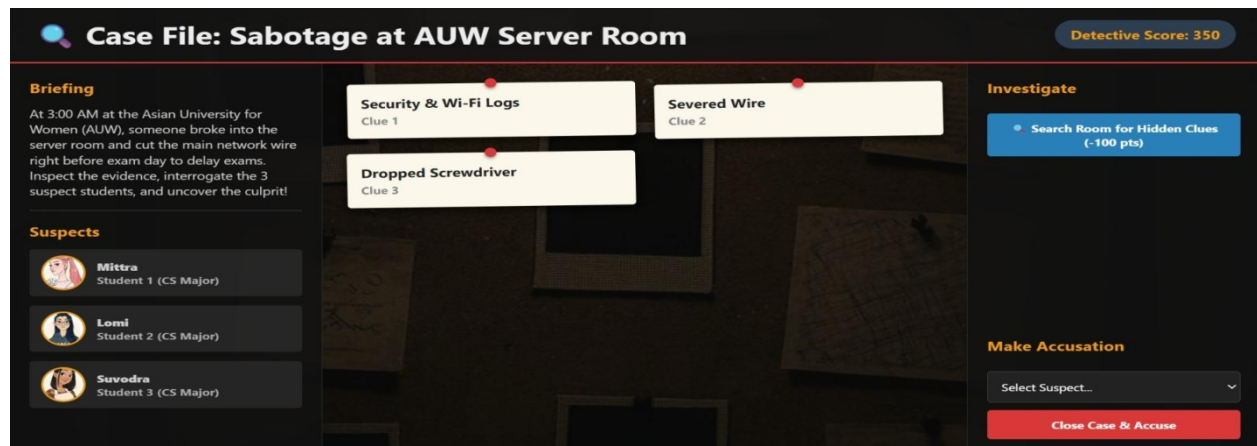
Limitations:

- **Session Persistence:** Reloading the web browser resets the detective score back to 350 points and clears injected hidden cards.
- **Fixed Narrative Path:** The current implementation features a single culprit outcome (Mittra).

6. Conclusion

- The "Sabotage at AUW Server Room" project successfully demonstrates how HTML, CSS, and JavaScript interact to build an engaging, state-driven web application. The game seamlessly handles dynamic score deductions (-100 pts for hidden clue searches and -80 pts for false accusations), evidence modal overlays, dynamic DOM injection, and game-ending victory logic.

7. Individual Contributions



1. Mittra - HTML DOM Hierarchy & Layout

- **Specific Webpage Output Shown:**

- **Left Briefing Panel:** The text container explaining the campus sabotage story and the 3:00 AM server incident.
- **Suspect Cards:** The vertical suspect list displaying round avatar portraits and student major labels for **Mittra**, **Lomi**, and **Suvodra**
- **Accusation Control Panel:** The bottom right dropdown selection box (`<select>`) labeled *"Select Suspect..."* and the red **"Close Case & Accuse"** submission button.
- **How Mittra's Work Shows on the Webpage:** Mittra authored the structural framework of `Final.html`. Her work provides the three-column DOM grid that holds the briefing text, suspect profile buttons, evidence card containers, and form inputs. Her layout binds user click events directly to the JavaScript modal triggers.

2. Lawmi - Visual Styling, Corkboard Layout & Dark Theme

- **Specific Webpage Output Shown:**
 - **Corkboard Visual Theme:** The dark wood/corkboard aesthetic (`bg-pic.jpg` overlay layer) visible behind all evidence cards.
 - **Evidence Cards & Red Pins:** The off-white rotated evidence cards featuring red circular pin icons (`.pin`) at the top center of each card.
 - **Modal Styling:** The dark grey modal dialog boxes with red border accents and clean white typography
- **How Lawmi's Work Shows on the Webpage:** Lawmi designed the full visual theme using CSS Grid and Flexbox. Her background-layer implementation allows the corkboard image to remain transparent without making text faint. She created the card hover effects, red pushpins on evidence items, custom button styling, and dark backdrop overlays (`rgba(0,0,0,0.7)`) that dim the screen when a clue is inspected.

3. Suvodra - Dynamic Logic, Score Deduction & Modal Engine

- **Specific Webpage Output Shown:**
 - **Score Badge Output:** The **"Detective Score: 350"** badge in the top right header (and its real-time updates to 250, 150, and 70).
 - **Hidden Clue Action Output:** The **"Search Room for Hidden Clues (-100 pts)"** button functionality
 - **Modal Pop-Up & Penalty System:** The **"Hidden Clue Found!"**, **"WRONG ACCUSATION!"** (-80 points deduction), and **"CASE SOLVED!"** pop-up modals
- **How Suvodra's Work Shows on the Webpage:** Suvodra programmed the state machine that calculates points and renders pop-ups. When a user clicks the search button, her script deducts 100 points from the header badge, dynamically injects a new evidence card onto the corkboard, and opens the modal pop-up overlay. When an incorrect accusation is submitted, her logic deducts 80 points and displays the warning modal.

